

CRMS Implementation Report

DBA_DEPLOYMENT and OBSERVATION Phases

Date: 2026-07-07

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Date: 2026-07-07
Repository: C:\Users\ADMIN\Desktop\crms_main

Overview

Integrated two new phases — DBA_DEPLOYMENT and OBSERVATION — into the CRMS application across backend, database, and frontend. The complete release lifecycle is now:

Draft !' RD Phase !' [RD Approval] !' FSD Phase !' [FSD Approval]
!' Development Phase !' Testing Phase !' UAT Phase
!' Deployment Phase !' [Deployment Approval]
!' DBA Deployment Phase !' Observation Phase !' Closed

1. Backend Controllers

1.1 src/controllers/moduleController.js

Change	Details
ALL_PHASES | Added DBA_DEPLOYMENT, OBSERVATION — now 9 phases
APPROVAL_PHASES | Left unchanged at DRAFT, RD, FSD, DEPLOYMENT — new phases are NOT approval phases
ensurePhaseGroupRows() | Added: auto-creates missing CRMS_PHASE_GROUPS rows for any phase not yet in the table. Called from getAll().
ensurePhaseSchema() | Added then removed: originally updated CHECK constraints at startup. Removed per user request.

1.2 src/controllers/approvalController.js

Change	Details
AFTER_APPROVAL_DEPLOYMENT | Changed from mapping to Closed to DBA Deployment Phase
phaseToNext.DEPLOYMENT | Now maps to DBA_DEPLOYMENT
phaseCodeToState | Added DBA_DEPLOYMENT: DBA Deployment Phase

1.3 src/controllers/releaseController.js

Change	Details
MANUAL_ADVANCE | Added DBA Deployment Phase: Observation Phase
stateToPhaseCode() | Added DBA Deployment Phase: DBA_DEPLOYMENT and Observation Phase: OBSERVATION
phaseToState() | Added DBA_DEPLOYMENT and OBSERVATION entries
phaseLabel() | Added DBA_DEPLOYMENT and OBSERVATION label entries
advanceState() phase map | Added DBA Deployment Phase: DBA_DEPLOYMENT
Observation Phase handling | Blocked advance — returns error "Use Close CR to move from Observation to Closed"
SEND_BACK_MAP | Added DBA Deployment Phase: Deployment Phase and Observation Phase: Development Phase
closeCR() | New endpoint: only available in Observation Phase, closes the CR
writeStateChange() | Added state-to-phase mapping; auto-assigns CR Owner as sub-task assignee in Observation Phase
ensureReleaseTaskSchema() | Removed: previously dropped/recreated chk_rt_phase and chk_release_state constraints

1.4 src/controllers/taskListController.js

Change	Details
updatePhaseDate() state map | Added DBA Deployment Phase and Observation Phase entries
getLiveTaskList() stage map | Added DBA Deployment Phase: DBA_DEPLOYMENT and Observation Phase: UNDER OBSERVATION

2. Frontend (cr-management-system.html)

Location	Change
PHASE_OPTS | Added DBA_DEPLOYMENT, OBSERVATION
phaseLabels | Added DBA_DEPLOYMENT: DBA Deployment, OBSERVATION: Observation
phaseOrder | Added DBA_DEPLOYMENT, OBSERVATION
PHASE_LABELS2 | Added both new phase labels
PHASE_CODE_FROM_STATE | Added both new state-to-code mappings
ALL_ACTIVE_PHASES | Added DBA Deployment Phase (Observation excluded)
Phase modals | Added DBA_DEPLOYMENT and OBSERVATION entries
phaseLabelsMap | Added DBA_DEPLOYMENT: DBA Deployment Task
stateToPhaseCodeFE() | Added DBA Deployment Phase: DBA_DEPLOYMENT

3. Database Schema (SQL)

3.1 CHECK Constraints Updated

Table	Constraint	New phases added
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crms_phase_groups	chk_pg_phase	DBA_DEPLOYMENT, OBSERVATION
crms_phase_templates	chk_pt_phase	DBA_DEPLOYMENT, OBSERVATION
crms_approval_flows	chk_af_phase	DBA_DEPLOYMENT, OBSERVATION
crms_company_group_phase_map	chk_cgpm_phase	DBA_DEPLOYMENT, OBSERVATION
crms_release_tasks	chk_rt_phase	DBA_DEPLOYMENT, OBSERVATION
crms_releases	chk_release_state	DBA Deployment Phase, Observation Phase

3.2 SQL Files Modified (12 files)

CRMS_DDL_V2.sql, crms_v2_ddl.sql, crms_v2_ddl (1-3).sql, crms_v3_ddl.sql, crms_v3_changes.sql, crms_company_mapping.sql (+1), crms_observation_phase.sql, crms_reviewer_process_owner_changes.sql

3.3 New Migration Script

SQL/crms_add_dba_deployment_observation_phases.sql — 235-line idempotent script updating all constraints and inserting config rows for existing modules.

4. Startup Schema Modifications (Removed)

Removed	File	Details
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ensurePhaseSchema()	moduleController.js	Dropped/recreated 4 CHECK constraints on startup
ensureReleaseTaskSchema()	releaseController.js	Dropped/recreated 2 CHECK constraints + column additions
ensurePhaseSchema() call	server.js	Required and invoked ensurePhaseSchema after DB connect

The application now relies on the SQL migration script. No schema modifications occur during application startup.

5. Deleted Files

DBA_DEPLOYMENT.js (121 lines) — standalone frontend module merged into HTML.

6. Key Design Decisions

- DBA_DEPLOYMENT is a manual-advance (non-approval) phase like DEV/TESTING/UAT
- OBSERVATION is a terminal pre-close phase with no advance button
- APPROVAL_PHASES excludes both new phases (no approval gate)
- Phase group rows auto-created by ensurePhaseGroupRows() in getAll()